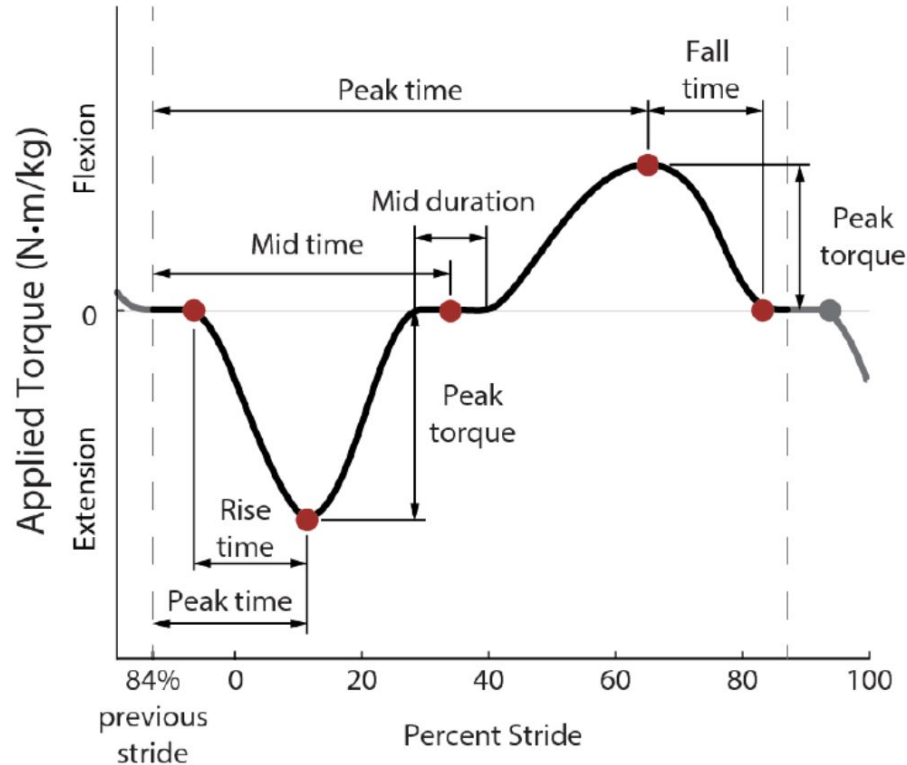


TBE & CNN-based Spline Controller GUI

Ryan Lee & Hridayam Tsunial

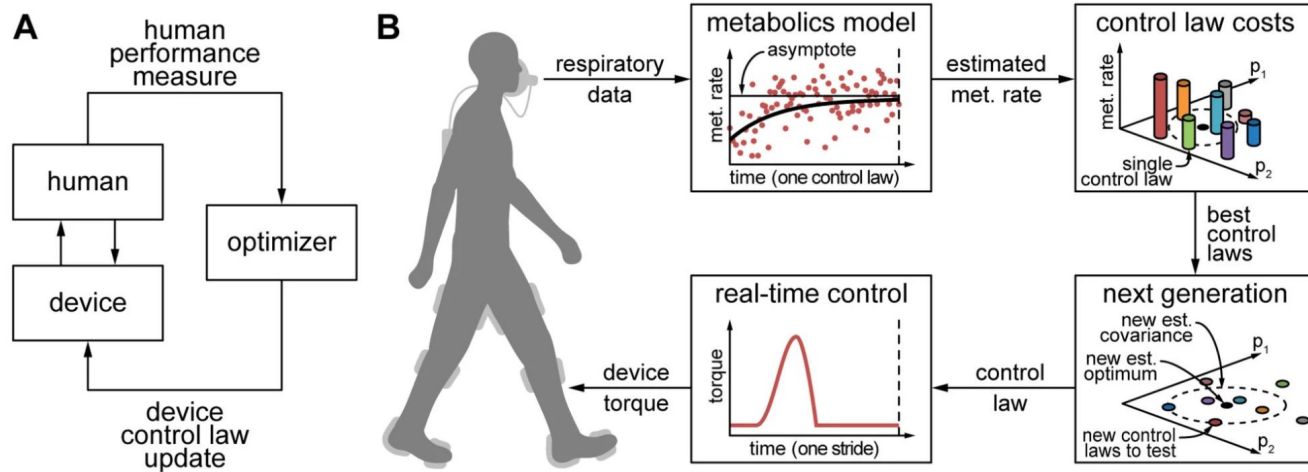
What is a Spline Controller?



Spline Controller

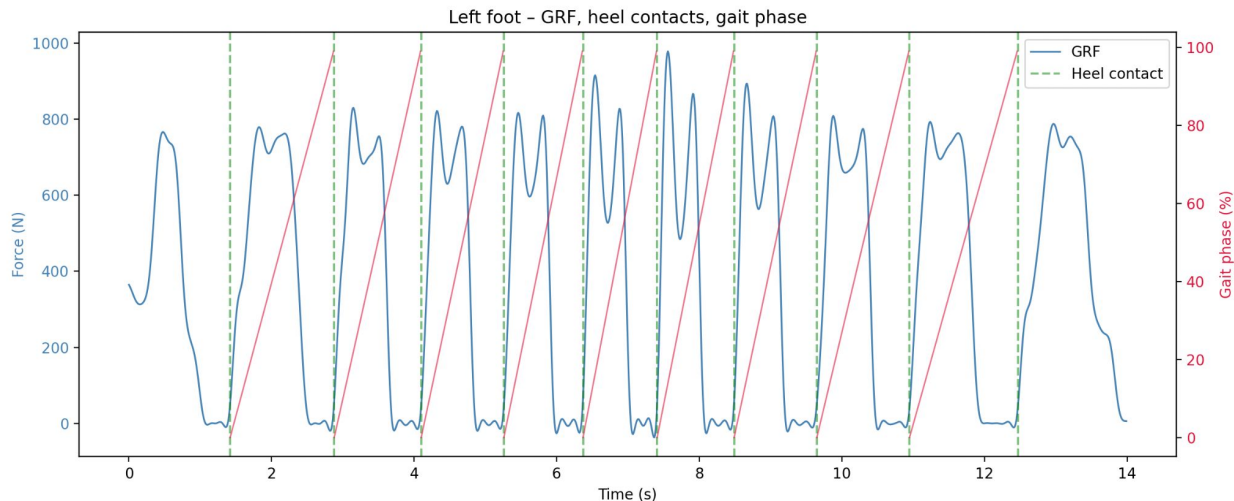
Adjustable control points (red dots) to define the torque profile

Using the Spline Controller

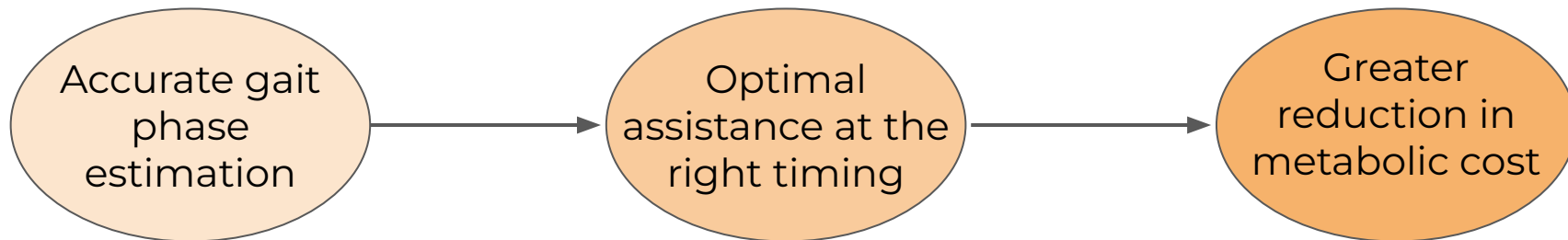


- Our assistance profile is based on Jimin walking on LG at 1.0 m/s
- If we want the most accurate torque assistance, we would need every subject to go through this process

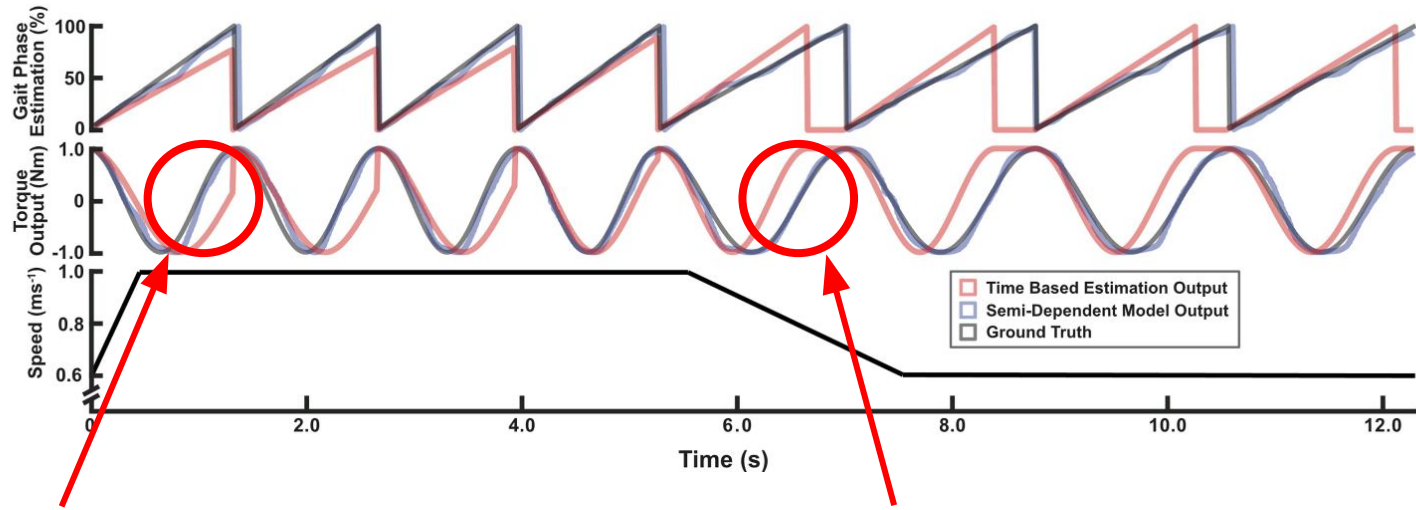
Gait Phase Estimation (GPE)



Left foot ground reaction force (GRF) & gait phases for a transient 15 second trial



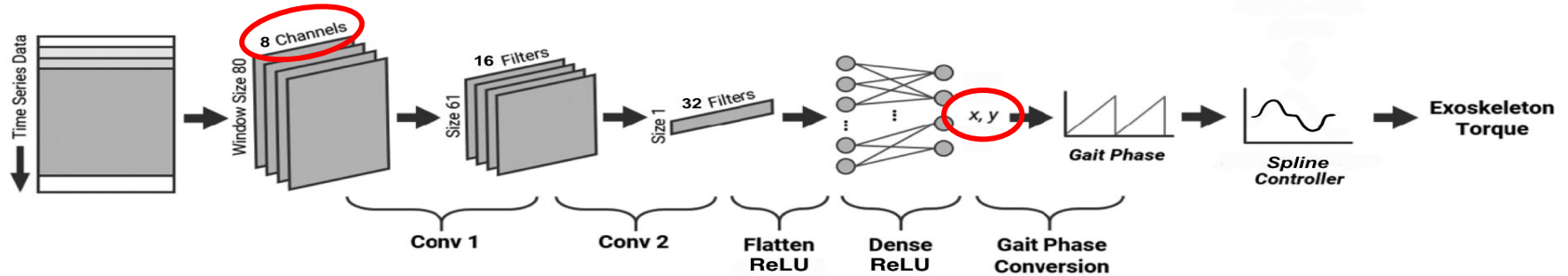
Time-Based GPE



Torque profile generated by TBE exhibits discontinuities

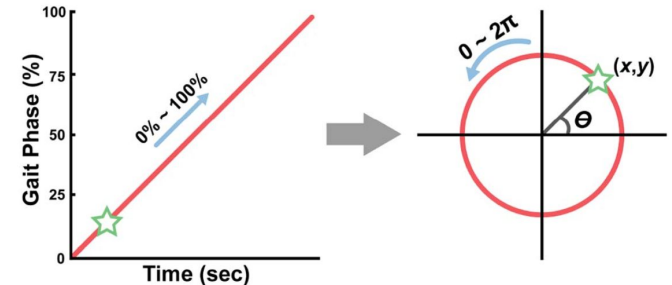
TBE output lags behind the ground truth curve when the speed changes

Developing CNN-based GPE

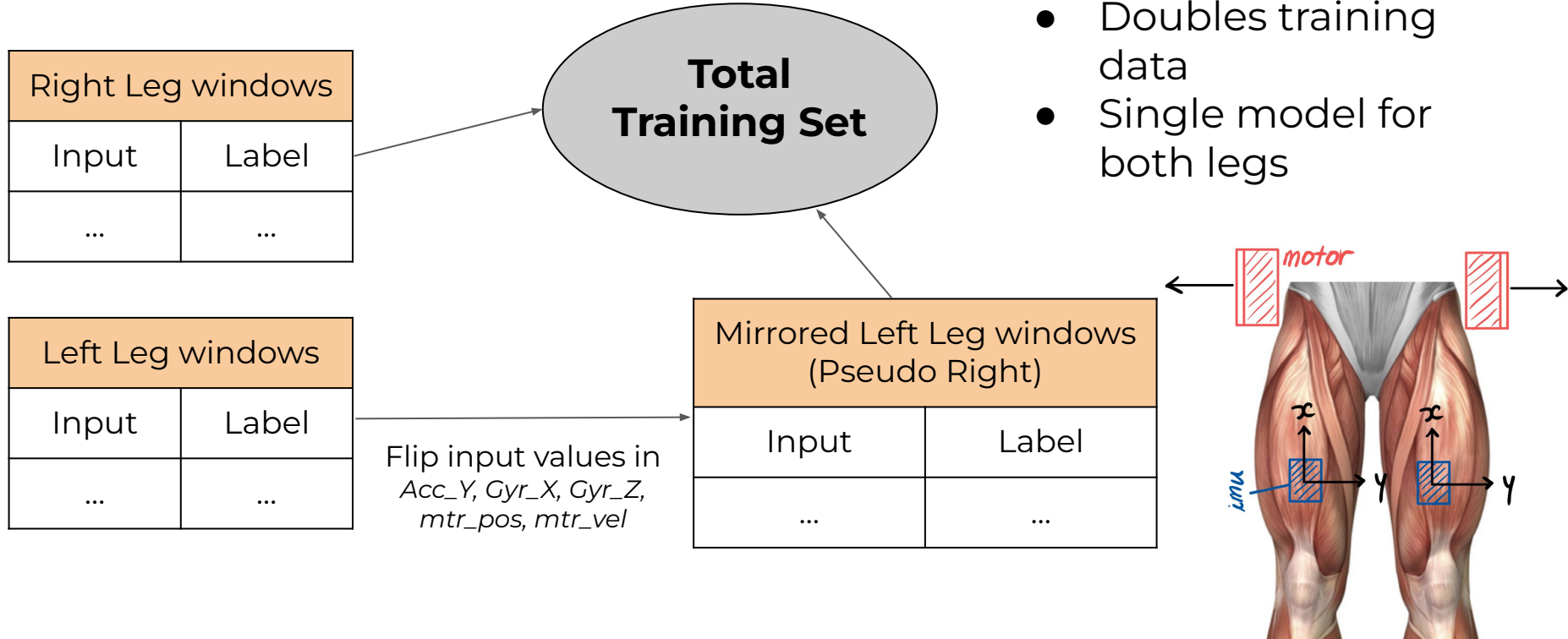


- **Input:** 8 channel data (Thigh IMU: Acc & Gyr $[x, y, z]$, Motor Encoder: pos, vel)
- **Label / Output:** x & y cartesian coordinates

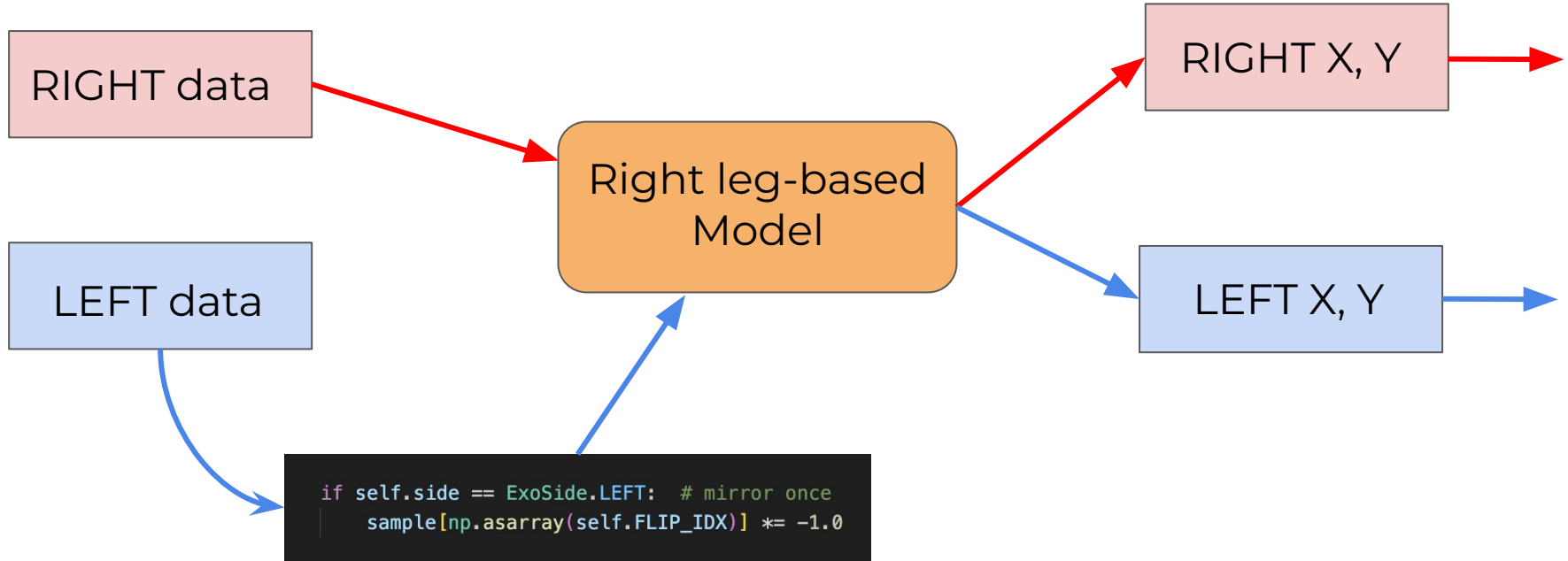
$$gait\ phase = \left(\left(\tan^{-1}\left(\frac{y}{x}\right) + 2\pi \right) \bmod 2\pi \right) \times \frac{100}{2\pi}$$



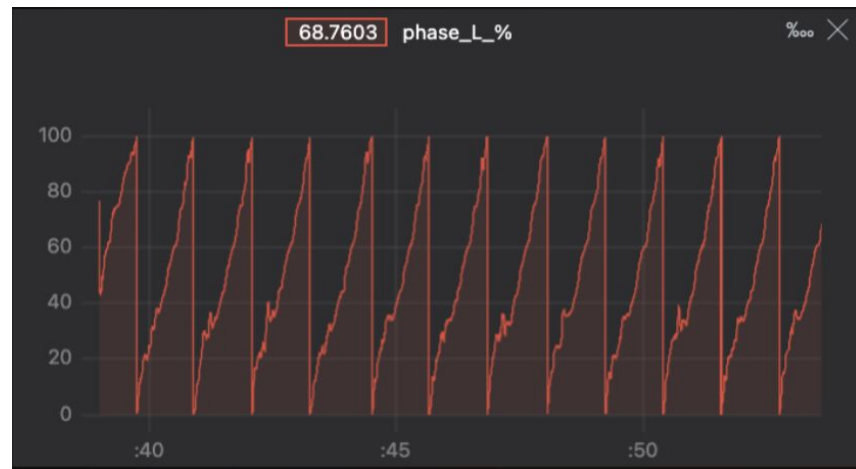
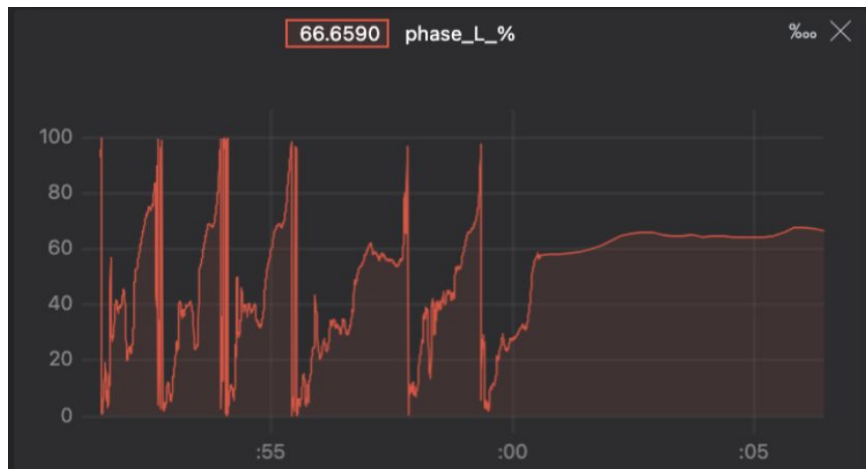
How to train a Unilateral Model



Real-Time Implementation



CNN Parameters



Left gait phase plots before and after changing parameters

- **Window size** = 40 → 80
- **Kernel size:**
 - 1st Conv layer = 5 → 20 (local kernel)
 - 2nd Conv layer = 5 → 61 (global kernel = window - local_k + 1)

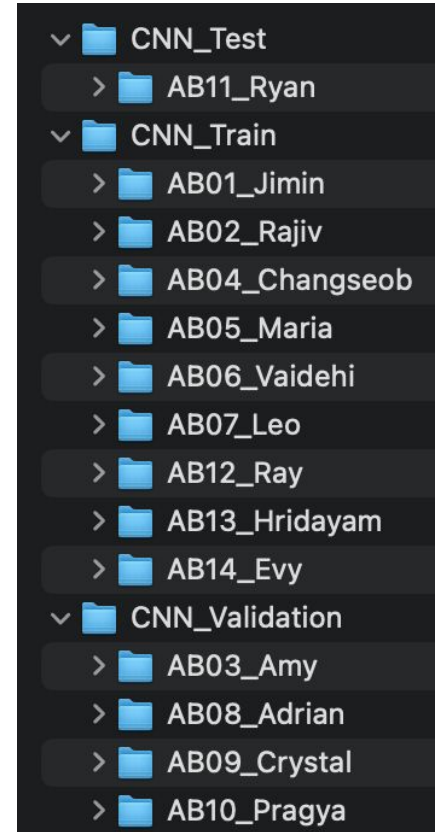
Types of Models Developed

Subject-Independent Models (Ryan, Hridayam):

- N=14 Data → Train: 9, Validation: 4, Test: 1
- Leave-one-subject-out (LoSO)
- Validation subjects chosen based on height & weight
 - Height bands: Low (< 170 cm), Mid (170 - 180 cm), High (> 180 cm)
 - Weight bands: Low (< 60 kg), Mid (60 - 75 kg), High (> 75 kg)

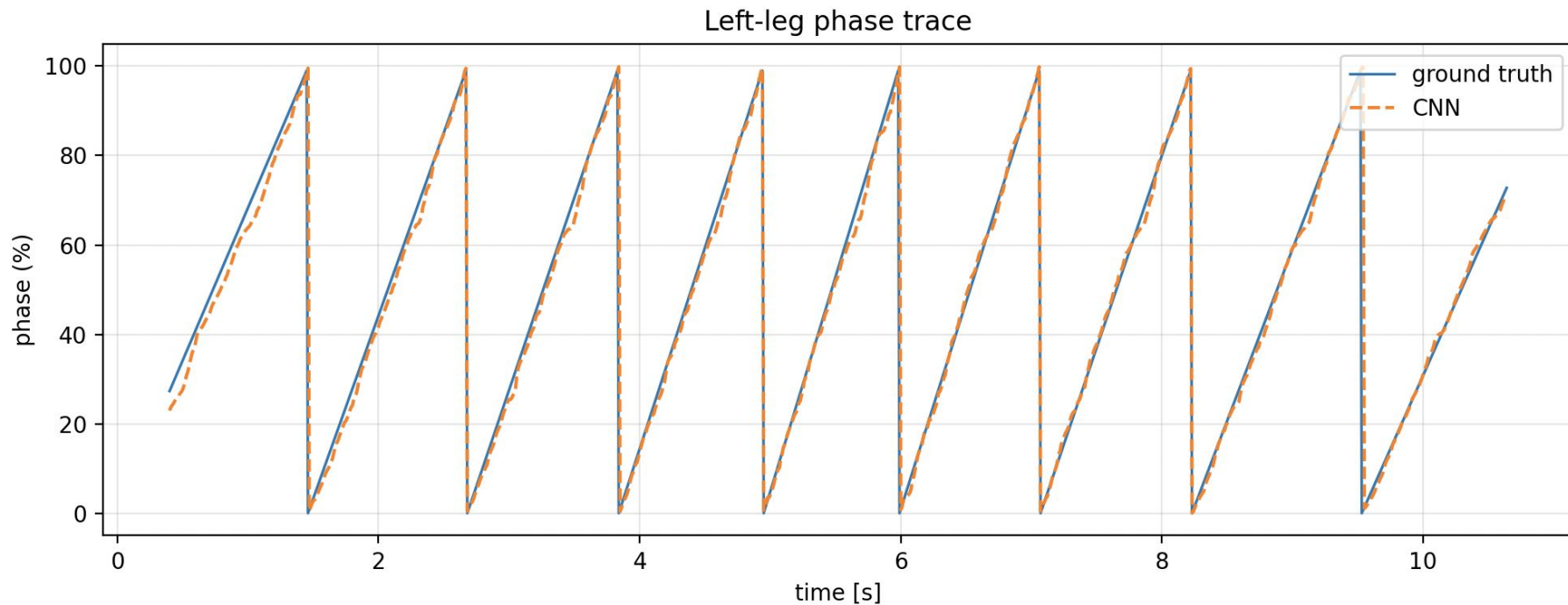
Synced Model & Unsynced Model:

- N=10 Data → Train: 6, Validation: 2, Test: 2
- Same subjects for each set
- Different data (Synced vs. Unsynced)



Distribution of data for IND_Ryan

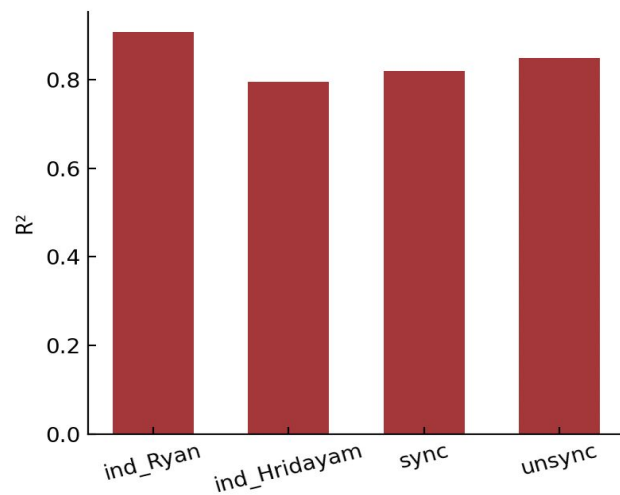
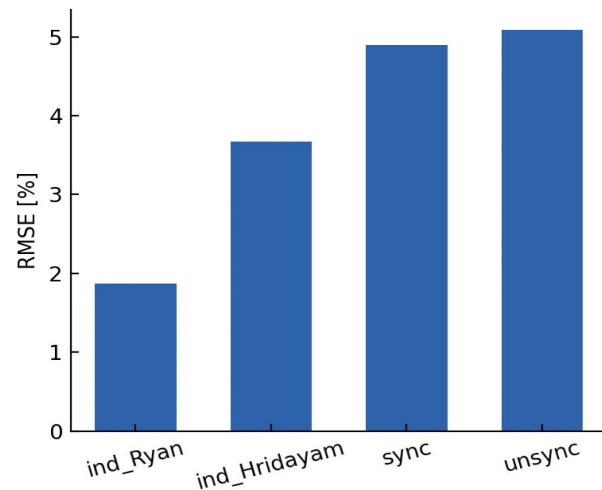
Offline Validations



Side-by-side trace of the predicted vs. ground truth left gait phases for Ryan's transient 15 second trial

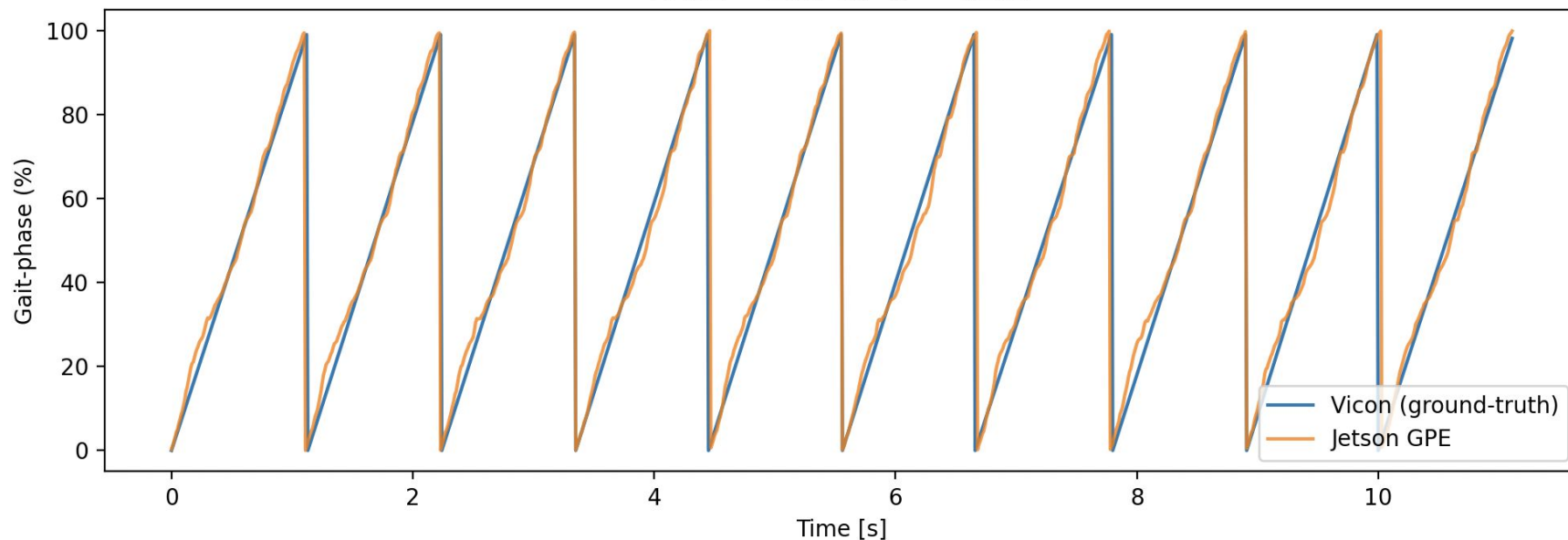
Offline Validation Results

Model	RMSE [%]	R2
IND_Ryan	ALL: 1.87 RIGHT: 2.00 LEFT: 1.74	ALL: 0.908 RIGHT: 0.916 LEFT: 0.904
IND_Hridayam	ALL: 3.67 RIGHT: 4.84 LEFT: 1.89	ALL: 0.795 RIGHT: 0.707 LEFT: 0.884
Sync	ALL: 4.90 RIGHT: 5.80 LEFT: 3.78	ALL: 0.820 RIGHT: 0.810 LEFT: 0.831
Unsync	ALL: 5.09 RIGHT: 6.23 LEFT: 3.60	ALL: 0.849 RIGHT: 0.807 LEFT: 0.890



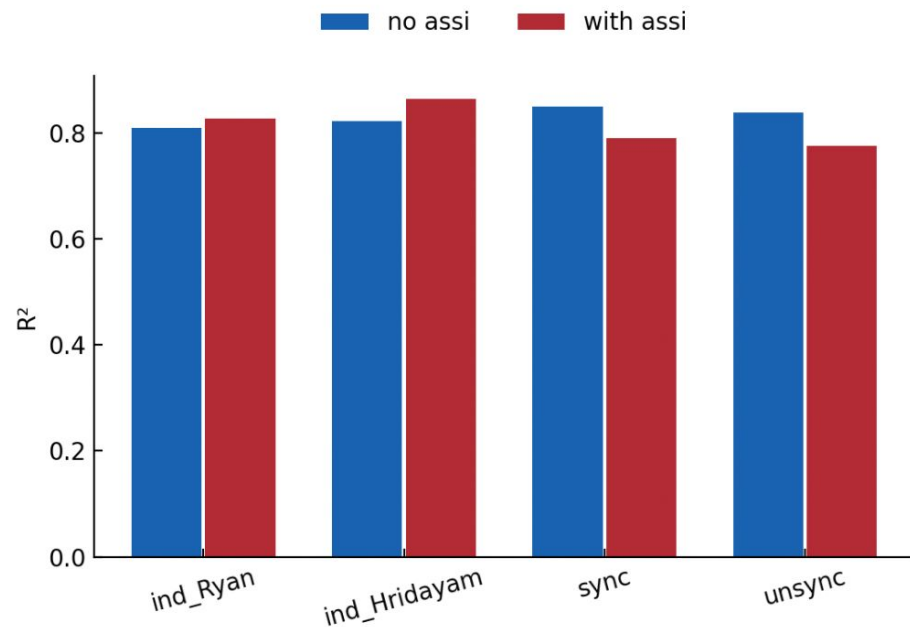
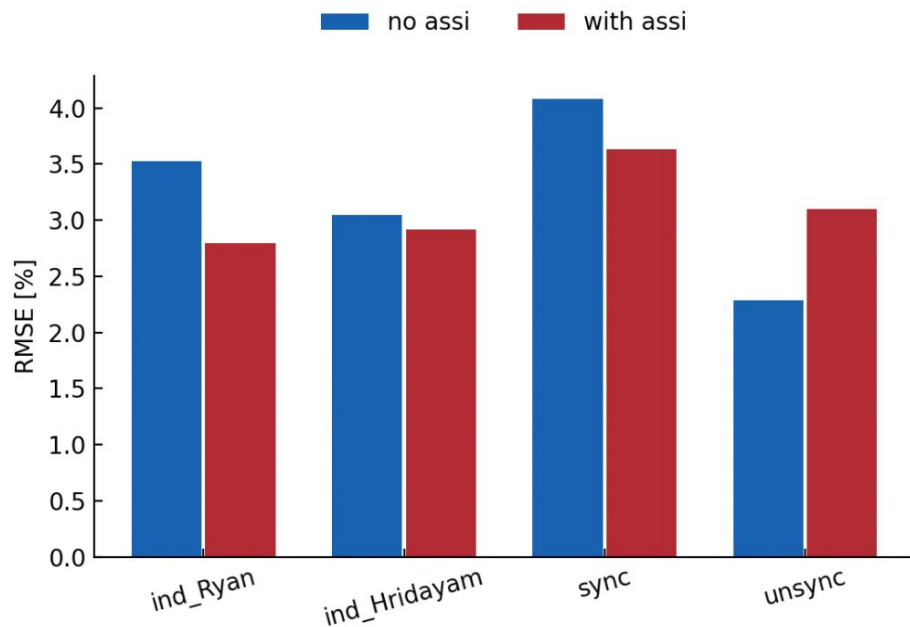
Online (Real-Time) Validations

Right - last 10 aligned cycles
RMSE = 2.68 %, $R^2 = 0.865$

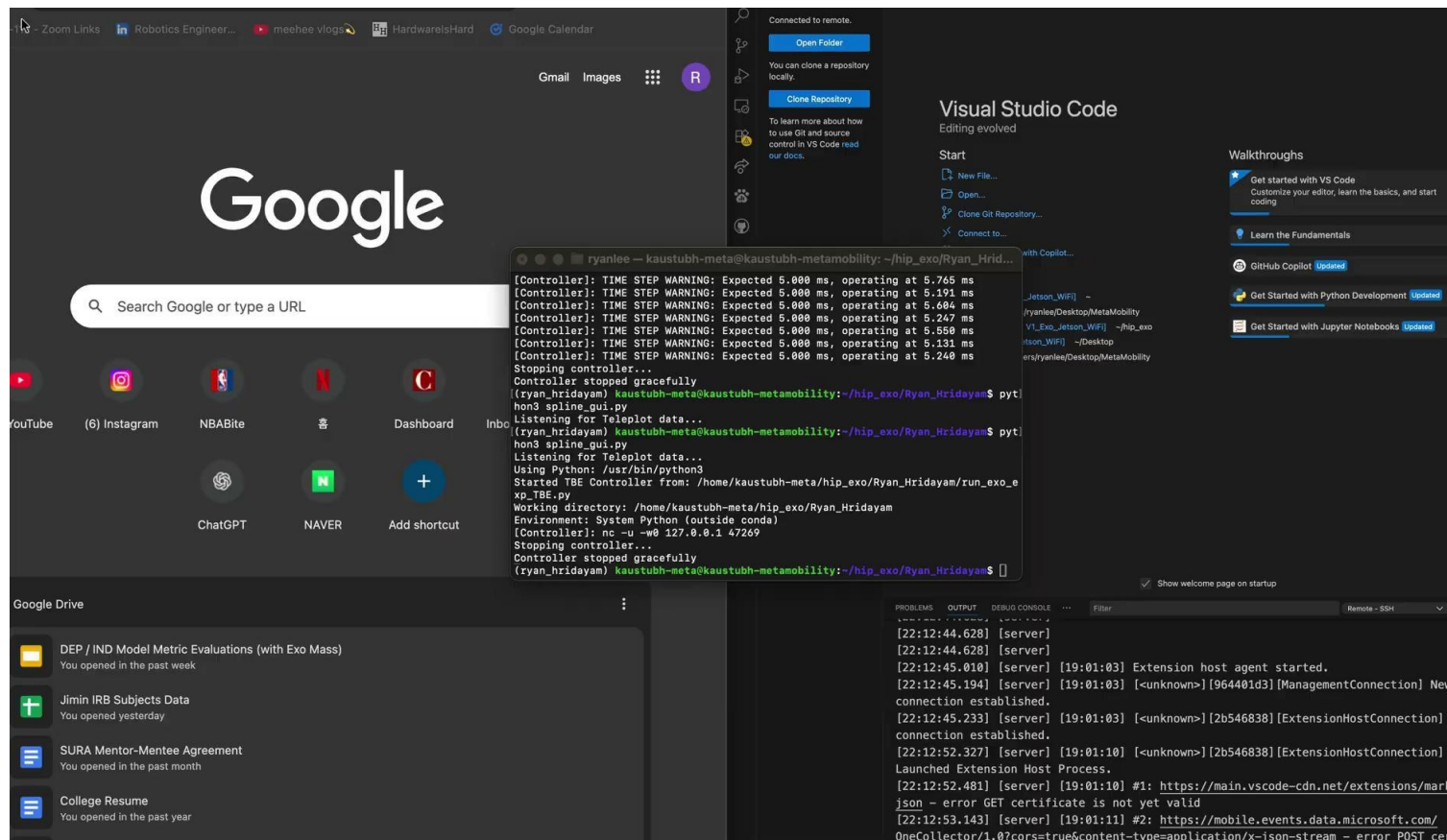


Side-by-side trace of the last ten left-leg strides collected in real time for IND_Hridayam with assistance

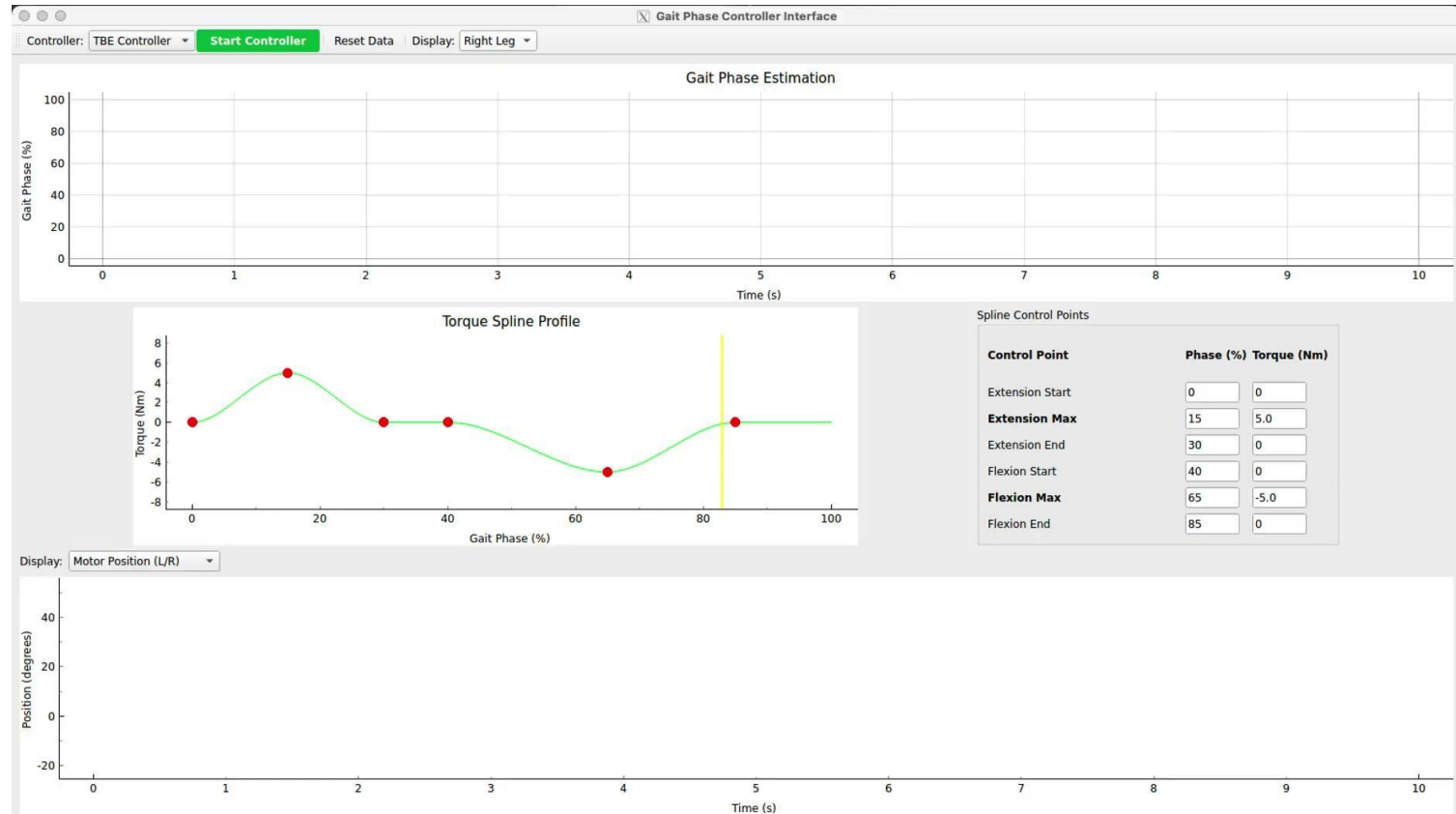
Online (Real-Time) Validation Results



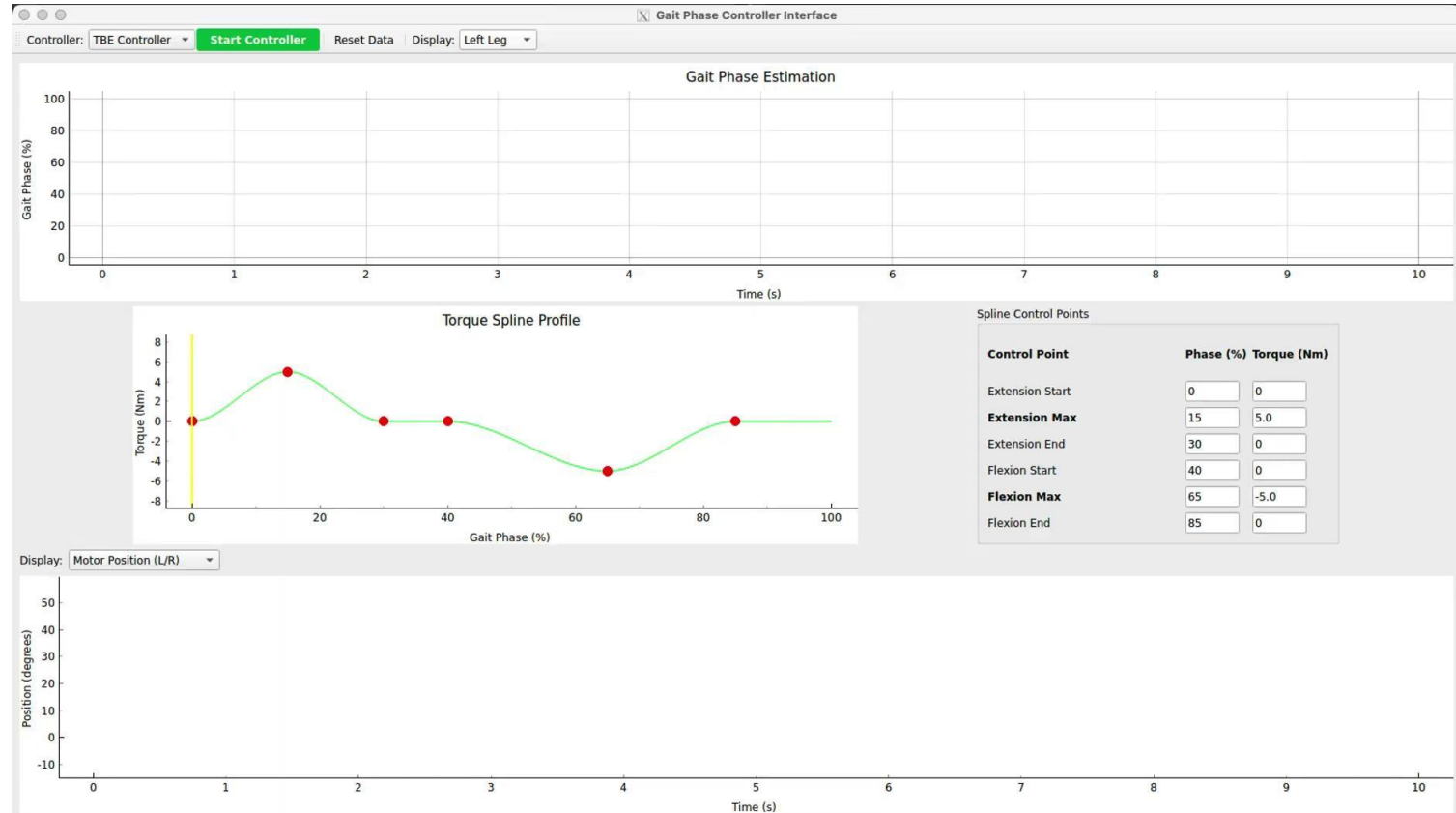
Spline GUI (with CNN GPE)



Spline GUI (with TBE GPE)



TBE (possible motor problem)



Future works

- Biotorque Controller GUI
- More accurately diagnosing motor problems and errors in left gait phase for TBE